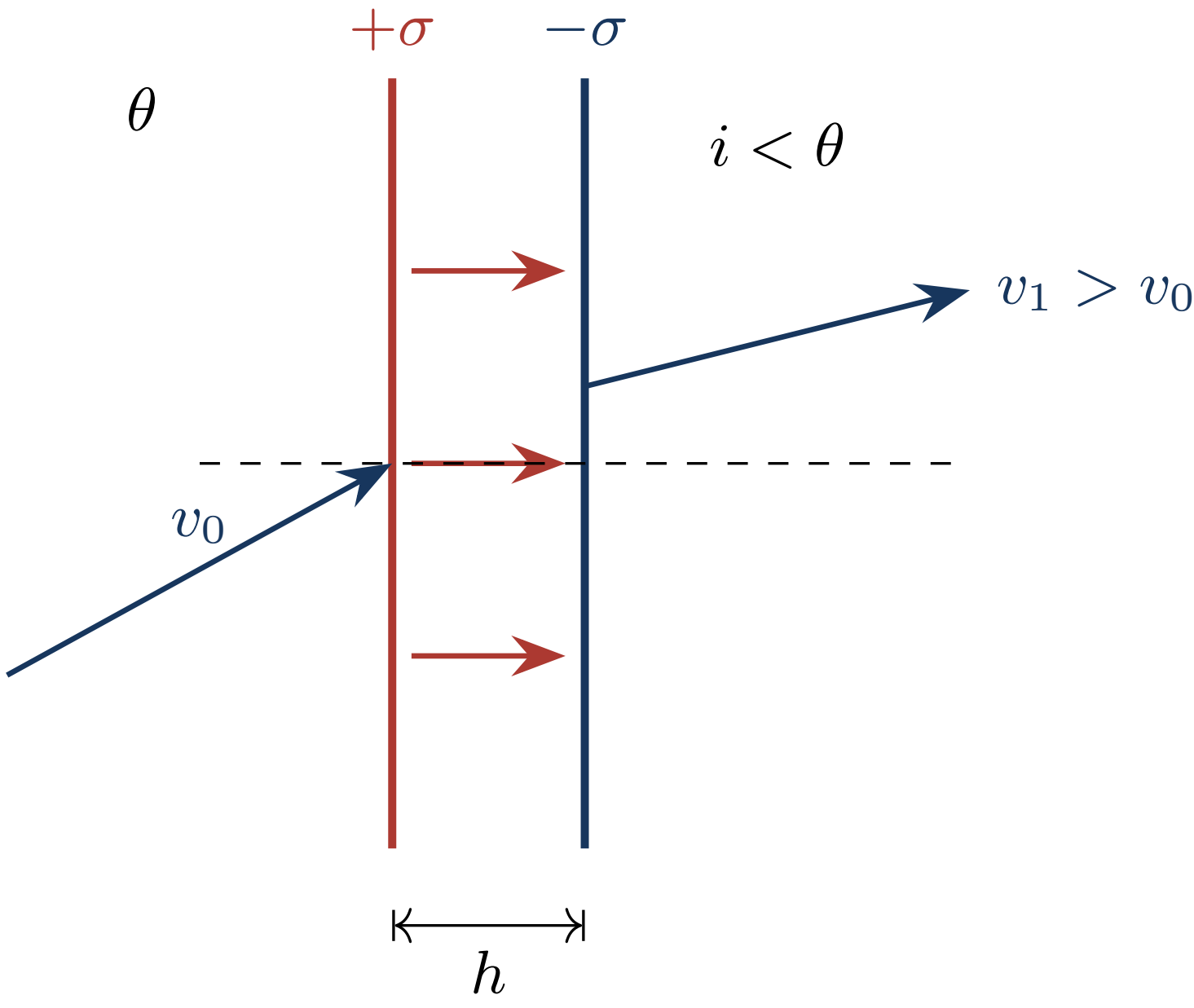
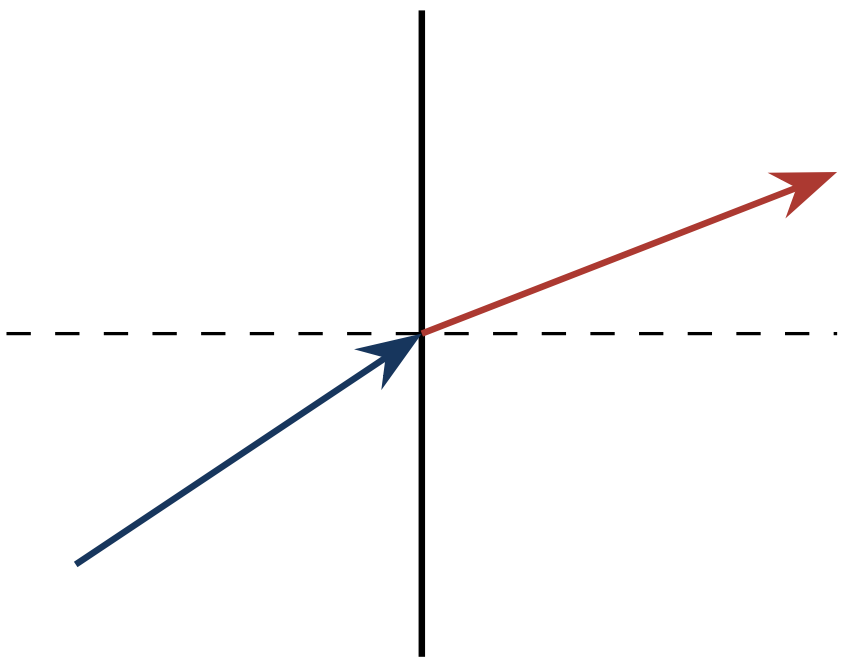


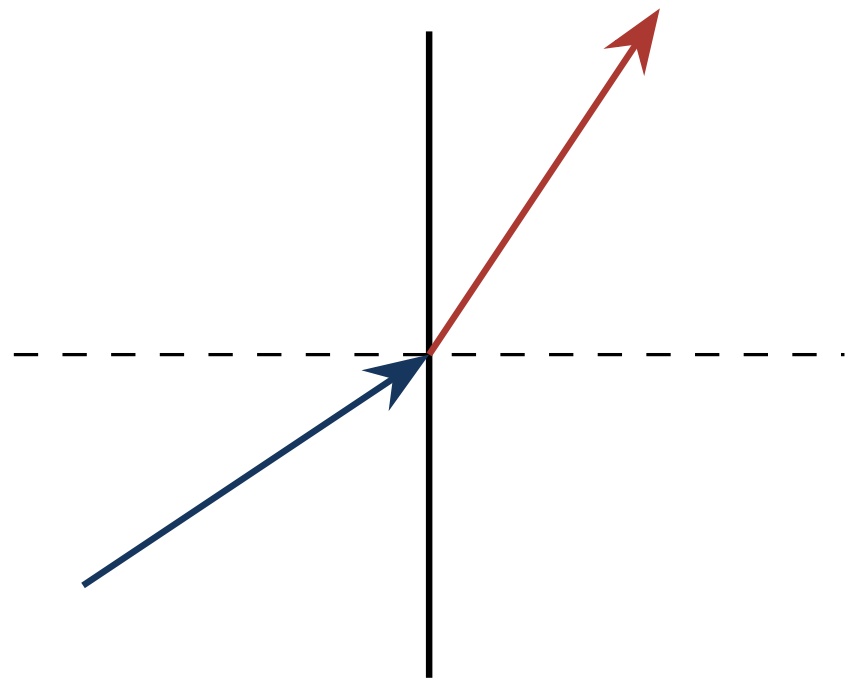
图示 $q > 0$：加速

加速



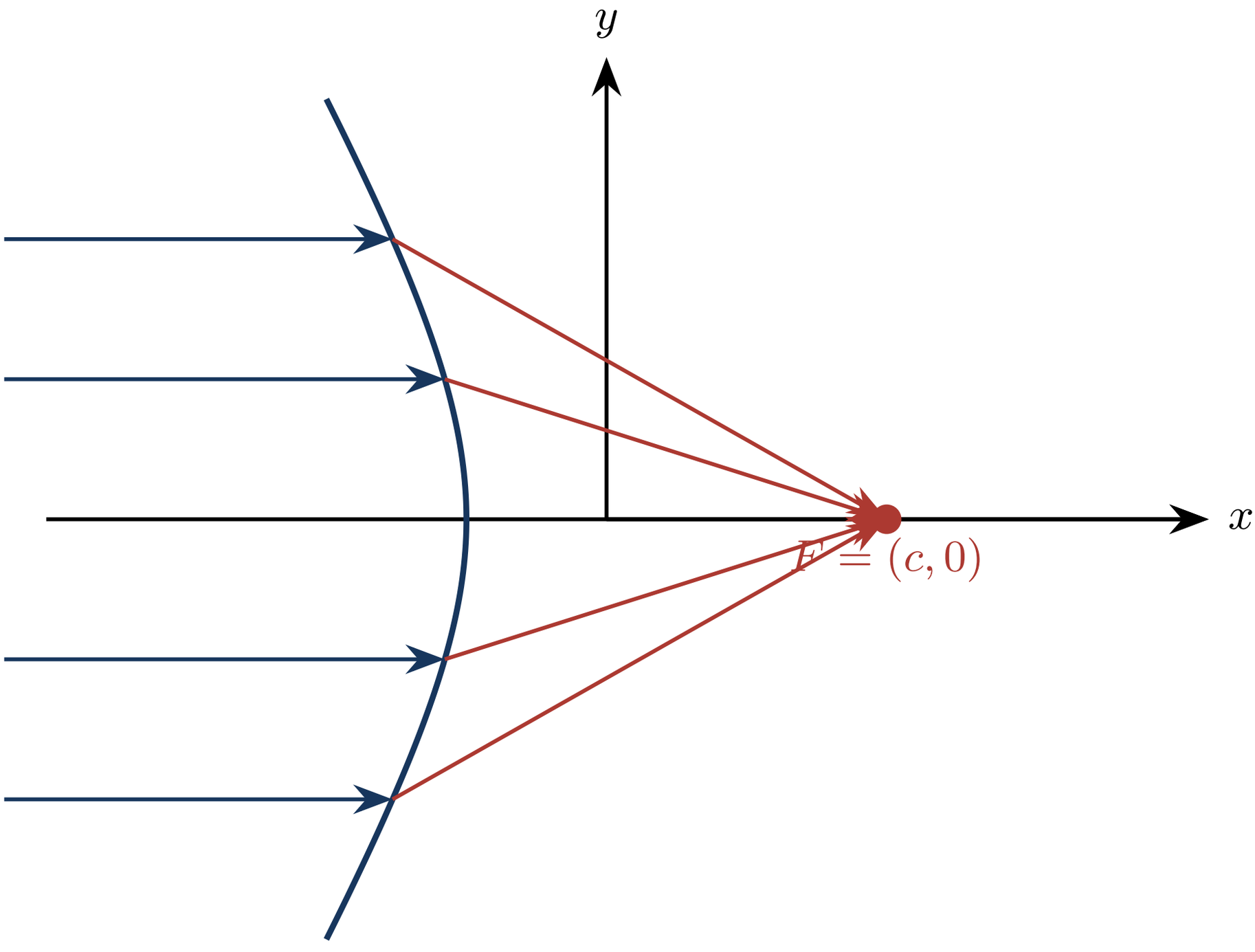
向法线偏折

减速

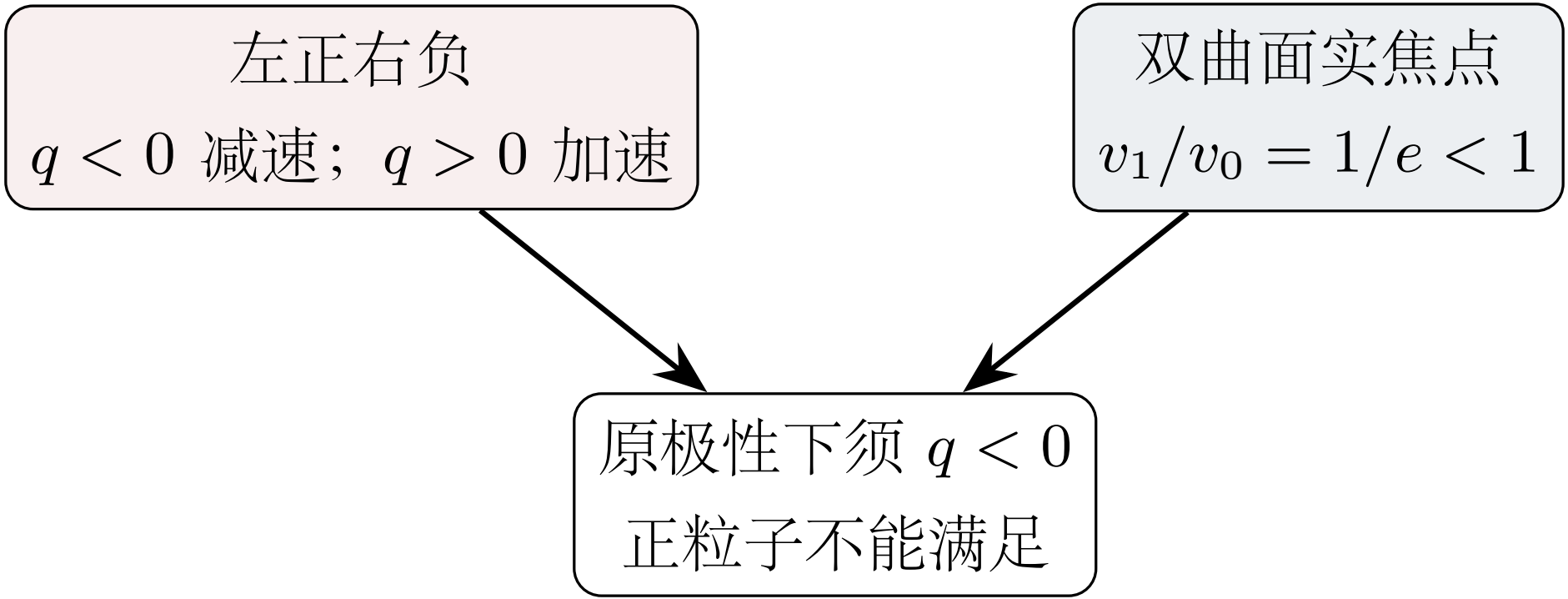


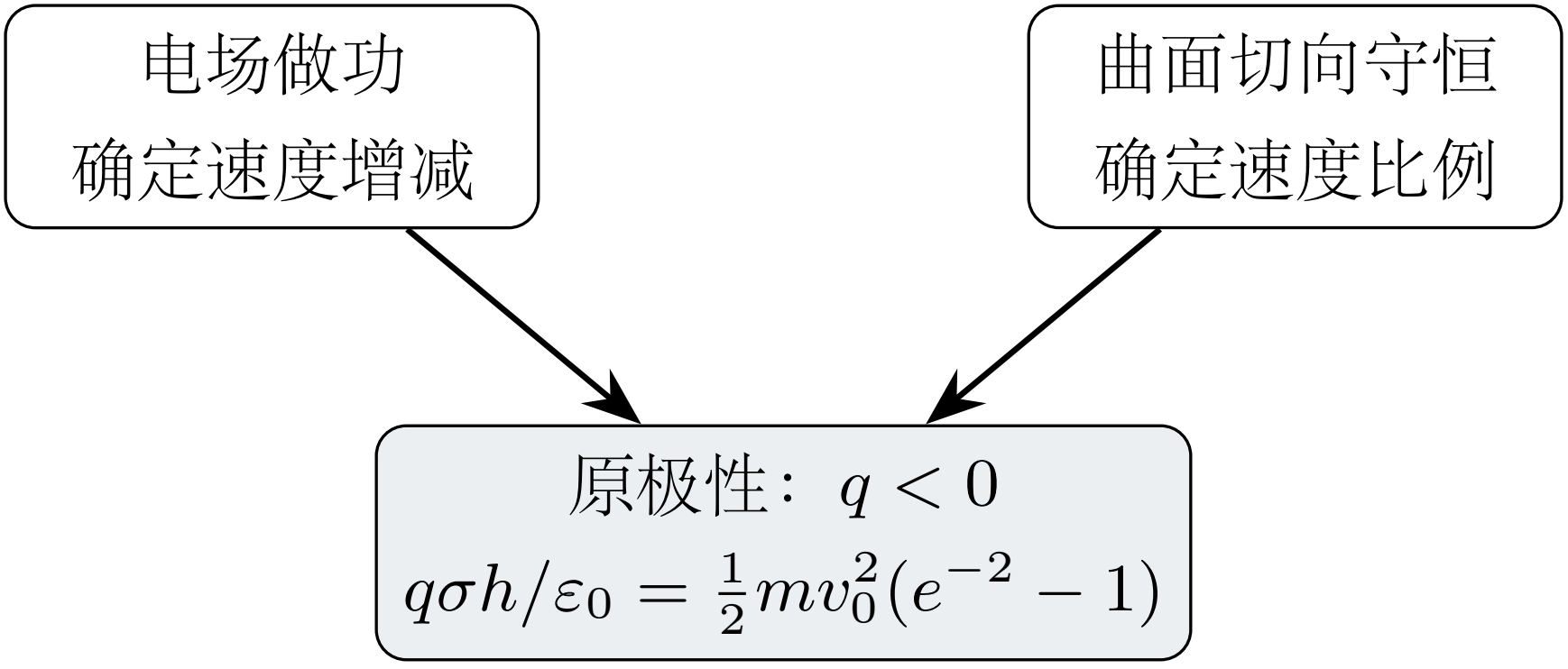
远离法线偏折

原极性：左正右负，$q < 0$ 减速



$$D = A - ex, \quad v_1 = v_0/e$$


左正右负
$q<0$ 减速；$q>0$ 加速
双曲面实焦点
$v_1/v_0=1/e<1$
原极性下须 $q<0$
正粒子不能满足



同时核查法向动能与薄层曲率近似